

1) Create table:

→ Creating a new table in the database.

Syntax:

CREATE TABLE tablename

(

column1 type1,

column2 type2,

);

Note: Here type1 and type2 in the syntax are the datatypes of column1 and column2 respectively.

→ ~~Database~~ ^{freely available} that are supported in SQL are mentioned below.

Data type	Syntax
1) integer	INT/INTEGER
2) Float	FLOAT
3) string	VARCHAR
4) Text	TEXT
5) Date	DATE
6) time	TIME
7) Datetime	DATETIME
8) Boolean	BOOLEAN

→ Boolean values are stored as integers 0 (FALSE) and 1 (TRUE).

→ Date object is represented as: 'YYYY-MM-DD'

→ Datetime object is represented as:

'YYYY-MM-DD HH:MM:SS'

1) Create a player table to store the following details of players.

column-name	data-type
name	VARCHAR(200)
age	INT/INTEGER
score	INT/INTEGER

⇒

```
CREATE TABLE Player (
    name VARCHAR(200),
    age INTEGER,
    score INTEGER
);
```

Note: We can check the details of the created table at any point in time using the PRAGMA Command.

PRAGMA: PRAGMA TABLE-INFO Command returns the information about a specific table in a database.
Syntax: PRAGMA TABLE-INFO (table-name);

Note: If the given table name does not exist, PRAGMA TABLE-INFO doesn't give any result.

Inserting Rows:

Insert clause is used to insert new rows in a table

Syntax:

INSERT INTO

tablename (column1, column2, ... columnn)

VALUES

(value1, value2, ..., valueN),

(value1, value2, ..., valueN),

(valueN, ..., valueN);

ex: INSERT INTO

Player (name, age, score)

VALUES

("RAKESH", 39, 35),

("Sai", 47, 30);

⇒ while inserting data, input value datatype should be same as the column datatype.

3) Retrieving Data:

→ SELECT clause is used to retrieve rows from a table.

→ Selecting specific columns: to retrieve the data of only specific columns from a table, add the respective column names in the SELECT clause.

Syntax:

SELECT

column1,
column2,
... ,
columnN

FROM

table.name;

★
ex:

SELECT
name,
age
FROM
Player;

→ attribute

name	age
Rakish	39
Sai	49

2) selecting all columns from a specific table

Syntax:

1) SELECT *
FROM table_name;

ex:

SELECT *
FROM Player;

3) Selecting specific rows.

→ We use WHERE clause to retrieve only specific rows.

Syntax: SELECT *
FROM table_name
WHERE condition;

→ WHERE clause specifies a condition that has to be satisfied for retrieving the data from a database

example:-

Get name and age of the Player table (only) whose name is 'Ram' from the Player table.
→ you can see only specific Row

SELECT *

FROM Player

WHERE name="sai";

→ output =

name	age
sai	49

4) Update rows in a table

UPDATE clause is used to update the data of an existing table in database. We can update all the rows or only specific rows as per the requirement.

1) Update all Rows

Syntax:- UPDATE

table_name

SET

columnname = Value;

★

ex:- update the age of all players to 100 in the Player table.

UPDATE

Player

SET

age = 100;

→ output

→

name	age
Trakesh	100
sai	100

2) Update Specific Rows:

Syntax: UPDATE

table_name

SET

columnname = Value

WHERE

columnname = Value;

ex: update the age of Ram Sai to 150 in the Player table.

⇒ UPDATE
Player
SET
Age = 150
WHERE
name = "Sai";

⇒

name	age
Sai	150
Raj	100

B) Delete Rows in a table

→ DELETE clause is used to delete existing records from a table.

1) Delete all Rows.

Syntax:-

DELETE FROM
table_name;

example:- Delete all rows from the Player table.

→ DELETE FROM
Player;

2) Delete specific Rows:-

Syntax:- DELETE FROM
table_name
WHERE
Column name = Value;

Example:-

Delete "Rajesh" from the Player table.

DELETE FROM

Player

WHERE

name = "Rakesh";

Note:- we can not retrieve the data once we delete the data from the table.

6) DROP Table

• DROP clause is used to delete a table from the database.

Syntax:- DROP TABLE tablename;

Ex: Delete Player table from the database.

DROP TABLE Player;

7) Alter Table:-

ALTER clause is used to add, delete, or modify columns in an existing table.

1) Add column in a table

Syntax:- ALTER TABLE
tablename

ADD

columnname datatype;

example:- Add a new column jersey_num of type integer to the Player table.

ALTER TABLE

Player

ADD

jersey_num INT;

Note: Default values for newly added columns in the existing rows will be null.

2) Rename column in a table

Syntax: ALTER TABLE

tablename RENAME COLUMN c1 to c2;

Example:

Rename the column jersey_num in Player table to jersey_number.

ALTER TABLE

Player RENAME COLUMN jersey_num TO jersey_number;

3) DROP COLUMN (Remove)

Syntax: ALTER TABLE

table_name DROP COLUMN column_name;

Example:

Remove the column jersey_number from the Player table.

ALTER TABLE

Player DROP COLUMN jersey_number;

Note:

DROP COLUMN is not supported in some DBMS, including SQLite.

SQL Concepts for freshers:- (next year ccbn 1-29)

- 1) Data, 2) Database 3) DBMS 4) Types of DBMS
- 5) create table 6) inserting rows 7) Retrieving Data 8) Update Rows 9) Delete Rows 10) ALTER
- 11) Comparison operators 12) string operators 13) logical operators 14) in and between operators 15) Order by and Distinct 16) Pagination 17) Group by 18) Group by with Having 19) SQL expressions 20) SQL functions 21) SQL Case 22) SQL set operators 23) Modeling database concepts 24) Joins 25) views & Subqueries 26) Transaction & Indexes.

Transaction control language (TCL) :- Commit, Roll back.

Data Definition language (DDL) :- Create, Alter, Drop, Truncate.

Data manipulation language (DML) :- Update, Insert, Delete, Select.

Data control language (DCL) :- Grant, Revoke.

Constraints: unique, not null, Primary key, foreign key, check.

NULL functions, triggers.

see sudha karthikesam Pdf SQL DBMS.

1) Data: Any sort of informations that is stored is called data.

Ex: messages & multimedia on whatsapp

- 1) Products and orders on Amazon
- 2) contact details in telephone directory, etc

2) Database:-

→ An organised collection of data is called a database.

3) Database Management system (DBMS):-

→ A software that is used to store easily and access data from the database in a secure way.

Advantages:-

- 1) security = data is stored & maintained securely.
- 2) ease of use
- 3) Durability & availability
- 4) Performance = quickly accessible to all the clients (applications and stakeholders).

4) Types of databases:-

→ There are different types of databases based on how we organized the data

1) Relation database (RDBMS) = In relational databases, the data is organised in the form of tables. (mysql, oracle, sqlite etc

2) Non-Relational Database:-

Non relational databases store the data in a 'non-tabular form'

ex: MongoDB, elastic search, Dynamodb, Cassandra etc

SQL

- SQL stands for structured query language.
- SQL is used to perform operations on Relational DBMS.
- SQL is declarative, Hence easy to learn

5) Create table:-

create table Command is using for creating new table

- PRAGMA TABLE_INFO command returns the information about a specific table in a database

6) Inserting Rows:-

INSERT clause is used to insert new rows in a table (existing)

7) Retrieving data:-

- SELECT clause is used to retrieve rows from a table

SELECTING SPECIFIC ROWS:-

→ WE USE WHERE clause to retrieve only specific rows.

→ WHERE clause specifies a condition that has to be satisfied for retrieving the data from a database.

8) UPDATE ROWS:-

UPDATE clause is used to update the data of an existing table in database.

→ UPDATE & SET combination.

9) DELETE ROWS:-

1) DELETE clause is used to delete existing records from a table.

DROP TABLE

DROP clause is used to delete a table from the database.

10) ALTER TABLE:-

→ ALTER clause is used to add, delete, or modify columns in an existing table.

11) COMPARISON OPERATORS:-

= equal to , <> not equal to , < less than ,

<= less than or equal to , > greater than ,

>= greater than or equal to

12) String Operators:

LIKE operator is used to perform queries on strings. This operator is especially used in WHERE clause to retrieve all the rows that match the given pattern.

13) Logical Operators:

→ AND, OR, NOT

NOT
AND
OR

↓
Precedence

14) IN and Between Operators:

1) Between 1000
AND 5000;

→ Between operator = Retrieves all the rows from table that have column value present between the given range (v_1 and v_2).

2) IN operator: Retrieves the corresponding rows from the table if the value of column is present in the given values ($v_1, v_2, \dots$).

eg brand IN ('Puma', 'Levi's', ...);

15) ORDER BY and DISTINCT:

→ We use ORDER BY clause to order rows. By default, ORDER BY sorts the data in the ascending order.

→ DISTINCT:

DISTINCT
distinct (f.e)

16) Pagination:

using page
can be set
request. And
be fetched

→ We use
select a

1) LIMIT
number
in result.

2) OFFSET
Position of
results

17) Aggregate:

Combining
is called

Aggregate

MAX, AV

Alias used
alternate
output.

→ DISTINCT :-

DISTINCT clause is used to return the distinct (i.e. unique values).

16) Pagination :-

Using pagination, only a chunk of the data can be sent to the user based on their request. And, the next chunk of data can be fetched only when the user asks for it.

→ we use LIMIT & OFFSET clauses to select a chunk of the results.

1) LIMIT clause is used to specify the number of rows(n) we would like to have in result.

2) OFFSET clause is used to specify the position from where the chunk of the results are to be selected.

17) Aggregation Functions :-

Combining multiple values into a single value is called aggregation.

Aggregate functions: COUNT, SUM, MAX, MIN, AVG.

Alias: Using the keyword AS, we can provide alternate temporary names to the columns in the output.

18) GROUP BY & GROUP BY WITH HAVING

→ GROUP BY clause in SQL is used to group rows which have same values for mentioned attributes

2) GROUP BY WITH WHERE:-

→ We use WHERE clause to filter the data before performing aggregation

3) HAVING: HAVING clause is used to filter the resultant rows after the application of GROUP BY clause.

Truncate:-

It will delete the complete records (data) except structure in the given table.

Syntax: Truncate table <tablename>;

DROP: It will delete records & structure also.

IN Operator:

The IN operator is used to check if a value is present in the list of values.

BETWEEN Operator:

BETWEEN operator is used to check if a particular value exists in the given range.

Note: IN and BETWEEN operators are used in where condition, etc.

WHERE V/S HAVING:

1) WHERE is used to filter rows and this operation is performed before grouping.

2) HAVING is used to filter groups and this operation is performed after grouping.

JOINS:

We use JOIN clause to combine rows from two or more tables, based on a specified column between them.

1) Natural join, inner join, full join, cross join, left join, Right join.

1) Natural join: Natural join combines the table based on the common columns.

2) Inner join: Combines rows from both the tables if they meet a specified condition.

3) Left join: in Left join, for each row in the left table, matched rows from the right table are combined. If there is no match, Null values are assigned to the right half of the rows in the temporary table.

4) RIGHT JOIN :-

Right join or right outer join is vice versa of left join.

i.e. in Right join, for each row in the right table, matched rows from the left table are combined. If there is no match, Null values are assigned to the left half of the rows in the temporary table.

5) Full join: Full join or full outer join is the result of both Right join & Left join.

6) CROSS JOIN:- in Cross join, each row from the first table is combined with all rows in the second table. (All possible combination)
Cross join is also called as Cartesian join.

7) Self

We can use
kind of
→ where
of a

~~STAR~~ PRIMA

→ The
that
→ The
should

→ A table
this
multiple

~~STAR~~ FOR

A foreign
Key or
→ foreign
→ the
Connect

7) Self join:

we can combine a table with itself, this kind of join is called self-join.

→ where joining condition is based on columns of a same table.

~~***~~ PRIMARY key:

- The Primary key is a column (or) set of columns that uniquely identifies each row in the table.
- The Primary key must contain unique values, should not be null and should never change.

Primary key = UNIQUE + NOT NULL

- A table can have only one Primary key and this Primary key can consist of single or multiple columns.

~~***~~ Foreign key:

A foreign key is a column points to the Primary key of another table.

- foreign key is used to create links b/w tables.
- the foreign key of one table helps to connect with the primary key of another table.

1) COMMIT:- It will make a transaction permanent.

2) RollBack: It will cancel any uncommitted transaction.

3) SavePoint: In b/w transactions we can define savepoint.

4) GRANT: It is useful to give the permission.

5) Revoke: It is useful to cancel the permission.

6) IsNull: It is useful to search for null values in a column.

is a column SQL $IS NULL$ AND $IS NOT NULL$

[inserting values multiple in a column new in a particular table.]

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Permanent

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$$100 + 100 + 100 + 100 = 400$$

[interesting values multiple in a column new in a particular table]